Case Study Report

Boris Boiarchuk

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1 Objective

The goal of this analysis is to design and evaluate a systematic intraday trading strategy across a universe of ≈500 equities, incorporating:

- realistic execution constraints,
- transaction costs,
- and a portfolio-level hedging approach.

The emphasis is on building a robust, transparent, and reproducible framework, rather than optimizing a specific strategy.

2 Data & Validation

We use five datasets:

- intraday prices (bid/ask, volume)
- ticker metadata (liquidity, borrow constraints, lot size)
- index constituents (weights)
- scheduled events
- fee schedule

Key validation steps

Before running any strategy, we perform strict data validation:

- remove duplicate bars (exact + (ticker, timestamp) conflicts)
- drop crossed markets (bid > ask)
- ensure positive bid/ask and valid timestamps
- validate metadata consistency (lot size, participation rates, borrow limits)
- ensure index weights sum to ≈1
- confirm ticker universe alignment

Rationale: Intraday backtests are extremely sensitive to data issues; incorrect spreads or timestamps can completely invalidate results.

3 Strategy Design

3.1 Signal: Rolling Mean Reversion

Based on what we from the data (mean-reversion with no visible trends, seasonal patterns or heteroskedasticity), exploiting the mean-reversion property seem to be the only sensible idea, so we use a rolling, stock-specific signal (details can be found in research.py file):

Rolling window: 100 observations ($\approx$ 100 minutes)

Compute:
• rolling mean
• rolling standard deviation
• Shift by 1 period to avoid look-ahead bias

Trading rules
• Long if: ask $\leq$ mean $-$ 1.0 $\times$ std
• Short if: bid $\geq$ mean + 1.0 $\times$ std

Interpretation: The strategy assumes short-term price deviations relative to recent history revert toward the mean.

3.2 Risk Filters

We suppress trading in two regimes:

Event window

• $\pm$10 minutes around scheduled events
• Rationale: avoid unpredictable volatility

Abnormal price zone

No trading if:

• ask $\leq$ mean $-$ 1.5 std
• bid $\geq$ mean + 2.0 std

Interpretation: Extreme deviations are treated as potential regime breaks rather than mean-reversion opportunities.

4 Portfolio Construction

• Total capital: \$10M
• Long book limit: 50%
• Short book limit: 50%
• Max per-name (per-ticker): \$25k

At each timestamp:

• capital is distributed equally across active signals
• positions are rounded to lot size
• short positions are capped by locate availability
This ensures:

• diversification,
• realistic sizing constraints,
• no over-concentration.

5 Execution Model

The execution engine simulates realistic constraints:

• Participation cap per name
• Volume-based trading limits
• Lot-size rounding
• Forced exits during:
· event windows
· abnormal zones

Important assumption: Forced exits may ignore participation limits to ensure risk control.

6 Transaction Costs

We include three cost components:

Explicit costs
• commissions
• exchange fees
• SEC fees (on sells)

Slippage
• proportional to spread:
· 25% of bid-ask spread × shares traded

Holding costs
• long financing cost
• short borrow cost (from metadata)

This makes the PnL much more realistic than a frictionless backtest.

7 Hedging Approach

We hedge using an index-weighted exposure proxy:

• Compute:
$\sum$ (**position notional** × **index weight**)

• Hedge by taking the opposite position in the index

Important clarification:
• This is not a true beta hedge
• It is a weight-based proxy hedge, due to lack of return-based beta estimates

We also:
• rebalance the hedge each minute
• apply the same cost model to hedge trades

8 Results

Key metrics reported

Performance
• Gross PnL
• Net PnL (after all costs)
• Hedged net PnL
• Sharpe-like metric (minute-based)
• Max drawdown

Trading behavior
• Total trades
• Win rate (after costs)
• Participation usage
• Forced exits

Risk & exposure
• average active names
• max gross notional
• sector concentration
• index exposure over time

Regime analysis
• event vs normal vs abnormal PnL
• signal bucket performance (by z-score)

9 Key Findings

Overall Performance
The strategy produces very large positive gross and net PnL, with net stock PnL of approximately $9.14B after all costs.

The extreme profitability is not evidence of a strong strategy, but rather a consequence of the synthetic data design: independent, stationary mean-reverting processes with large variance create arbitrage opportunities that would not persist in real markets.

Transaction costs (explicit fees + slippage) are relatively small compared to gross PnL:

• total trading costs: ≈$7.5M
• holding costs: negligible

This indicates that the simulated strategy exhibits an extremely strong apparent edge.

Signal Effectiveness
The rolling mean-reversion signal appears highly predictive:

- next-bar win rate (after costs): $\approx 59.4\%$
- Sharpe-like metric: ≈ 380

Interpretation:

- The signal generates consistent short-term directional accuracy
- Returns are not only positive but also very stable

However, the magnitude of performance suggests that:

- either the signal is very strong, or
- there may be structural assumptions in the simulation that inflate profitability (see limitations)

Trading Activity and Turnover
- total trades: ≈580k
- total shares traded: ≈114M
- total trade notional: ≈$18.3B

The strategy is high-frequency and high-turnover, but:

- costs remain small relative to PnL
- participation constraints do not appear binding on average

This suggests that:

- execution assumptions (especially slippage) are relatively favorable

Portfolio Construction & Diversification
- average active names: ≈286
- maximum active names: 384

The strategy is well diversified cross-sectionally, which:

- reduces idiosyncratic risk
- stabilizes returns

Sector concentration remains moderate:

- largest sector notional: ≈$197k on average

Risk Management & Regimes
- forced exits: ≈41k instances
- event and abnormal filters are actively engaged

These mechanisms:

- help control exposure during unstable periods
- likely reduce tail risk

Maximum drawdown is very small:

- ≈-$8.8k

This is extremely low relative to total PnL, suggesting very stable performance.

Hedging Effectiveness
- hedge PnL: slightly negative (≈-$443k)
- hedged Sharpe: nearly identical to unhedged
- exposure levels: relatively small

Interpretation:

- the portfolio already has low net index exposure
- the hedge provides minimal benefit, but incurs costs

This is consistent with:

- a largely market-neutral long/short construction
- and the use of an index-weighted proxy hedge rather than a beta hedge

Data Quality Observations

Validation identified:

- ≈3.4k duplicate rows
- ≈3.5k duplicate (ticker, timestamp) conflicts
- ≈1.25k crossed markets

All issues were:

- detected,
- logged,
- and cleaned prior to backtesting

This supports data integrity and reproducibility of results.

10 Conclusion

This project demonstrates a complete intraday trading framework, including:

- rigorous data validation
- a rolling mean-reversion signal
- realistic execution and cost modeling
- portfolio-level risk controls
- and a hedging overlay

The strategy produces strong simulated performance, with:

- high win rate,
- very large net PnL,
- and low drawdowns.

However, the magnitude and stability of returns suggest that results should be interpreted with caution.

In particular:

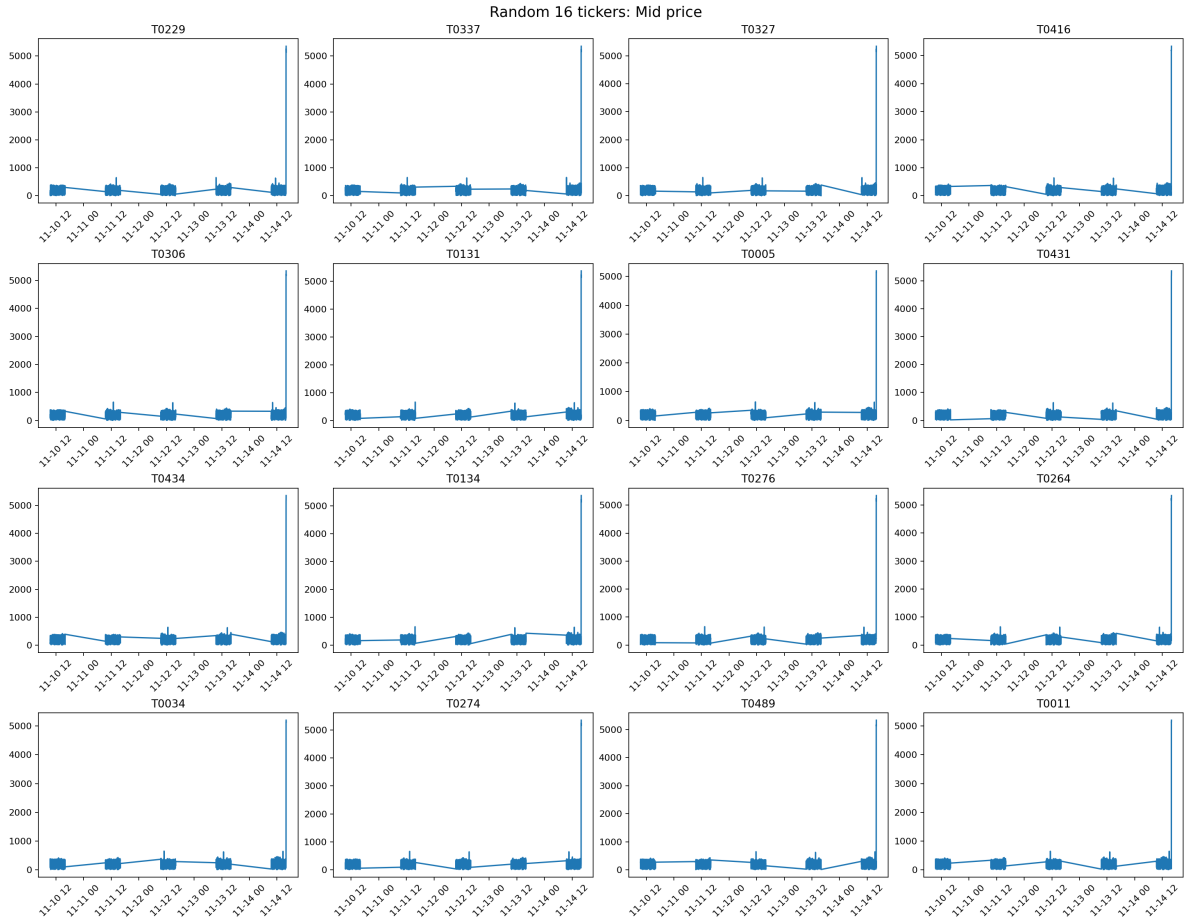
- the slippage model is simplified (spread-based)
- execution may be optimistic under high turnover
- the hedge is a proxy (not beta-based)
- and no parameter sensitivity or out-of-sample validation is performed

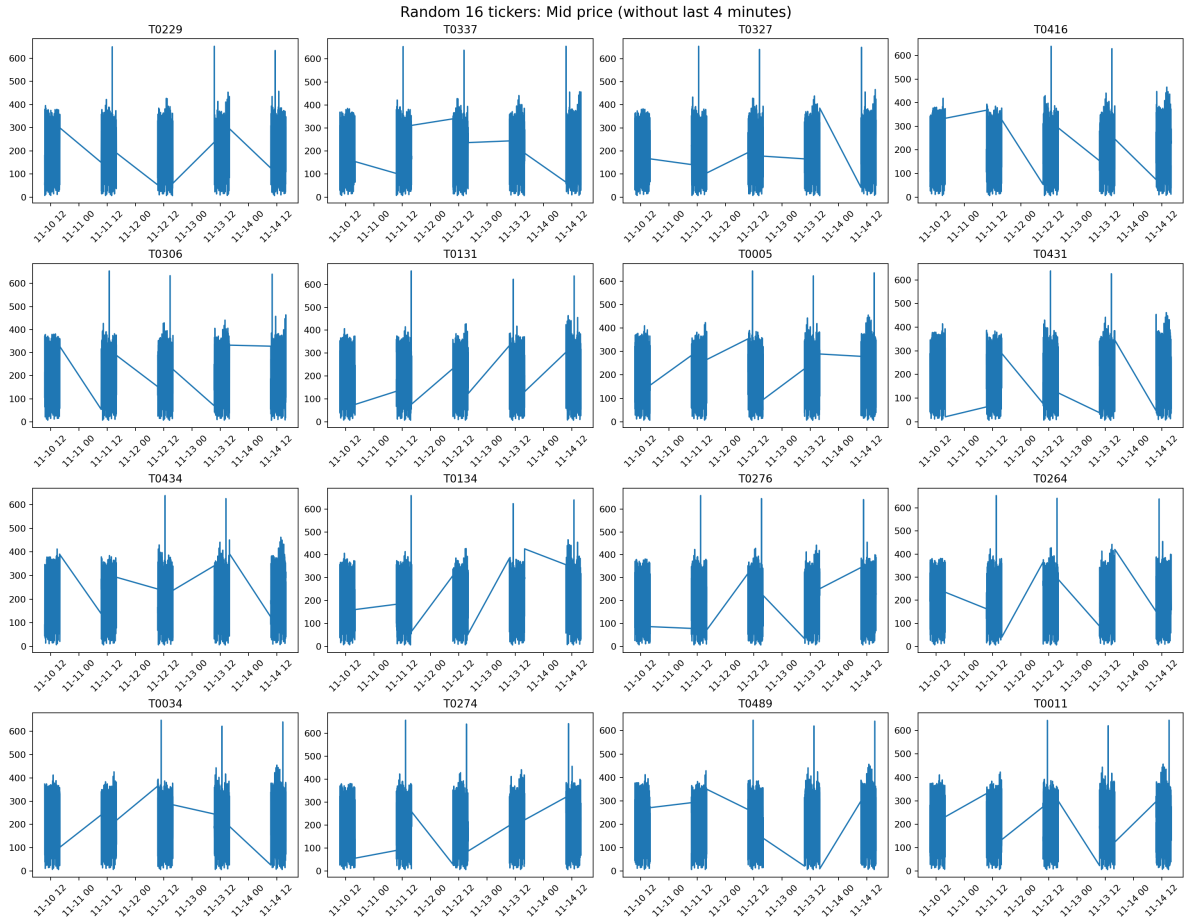
Therefore, the primary contribution of this work is:
a robust and transparent research framework, rather than a production-ready trading strategy.

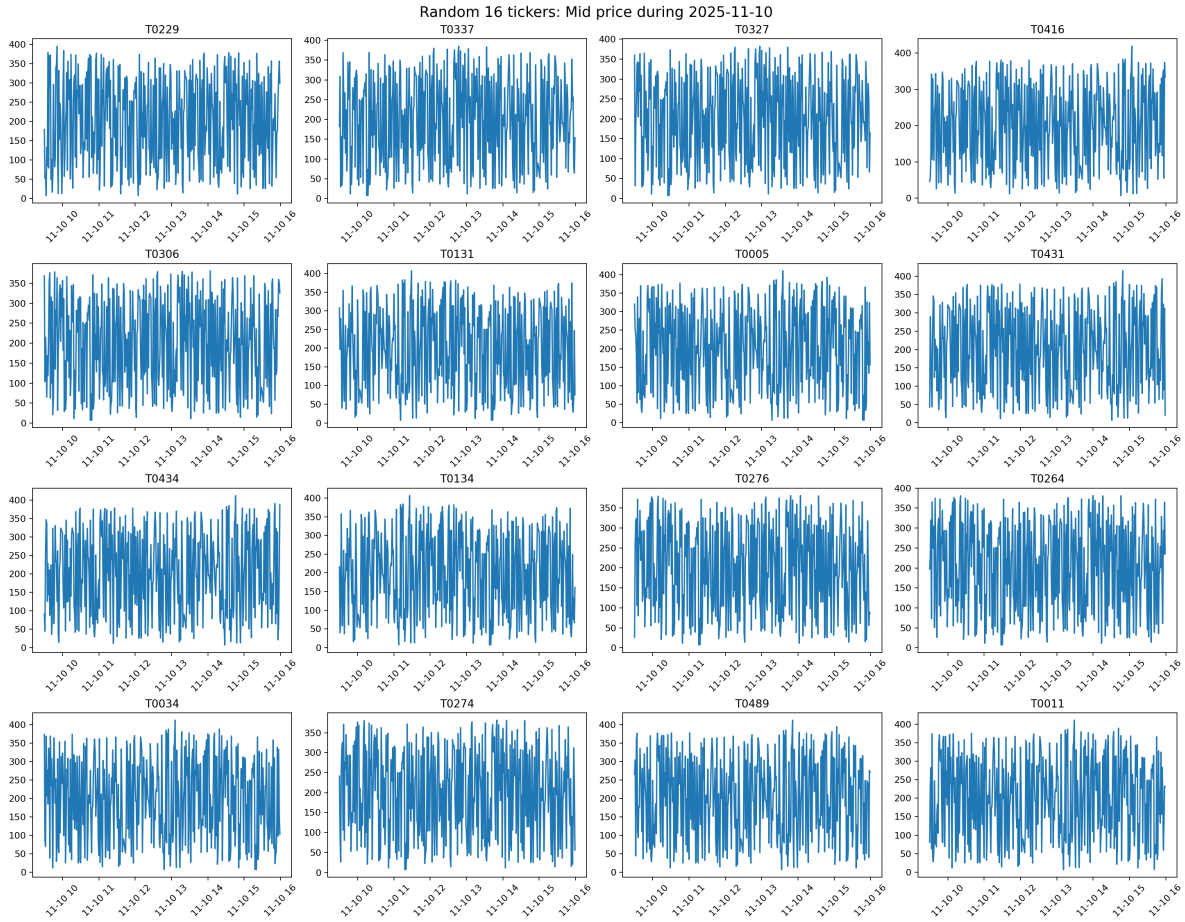
11 Appendix

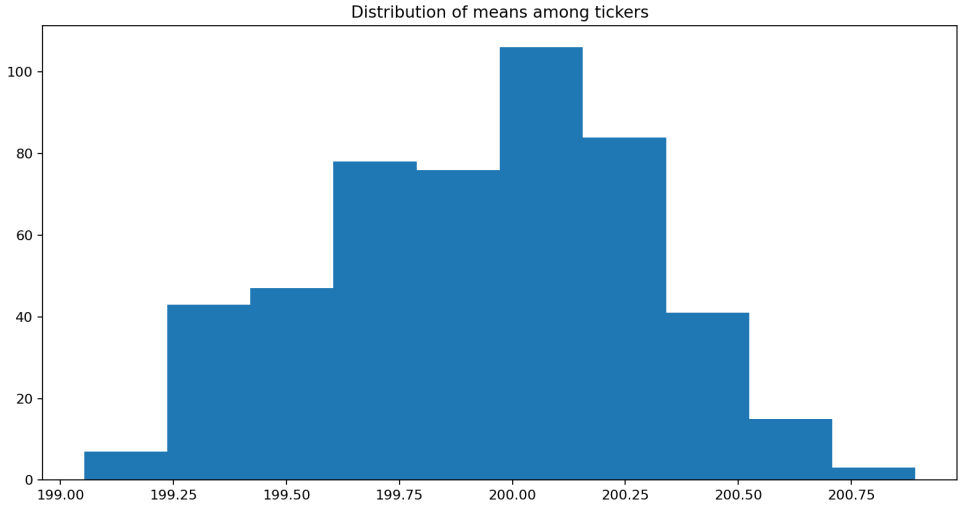
A. Data Cleaning Summary
- exact duplicates removed: 3,434
- duplicate key rows resolved: 3,496
- crossed markets removed: 1,250
- locked markets retained (monitored)

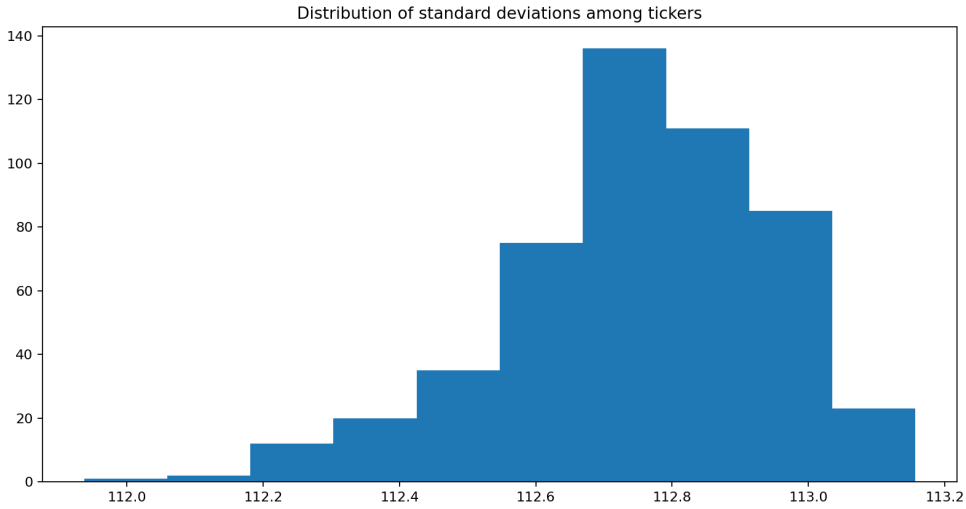
B. plots

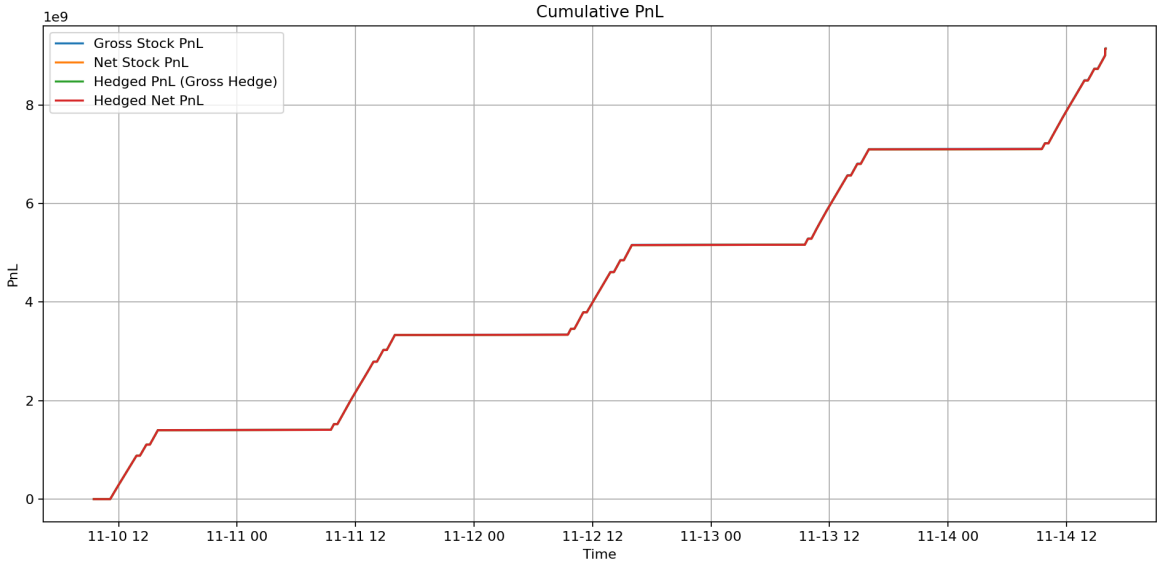

Random 16 tickers: Mid price
T0229
T0337
T0327
T0416
T0306
T0131
T0005
T0431
T0434
T0134
T0276
T0264
T0034
T0274
T0489
T0011
0
1000
2000
3000
4000
5000
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12

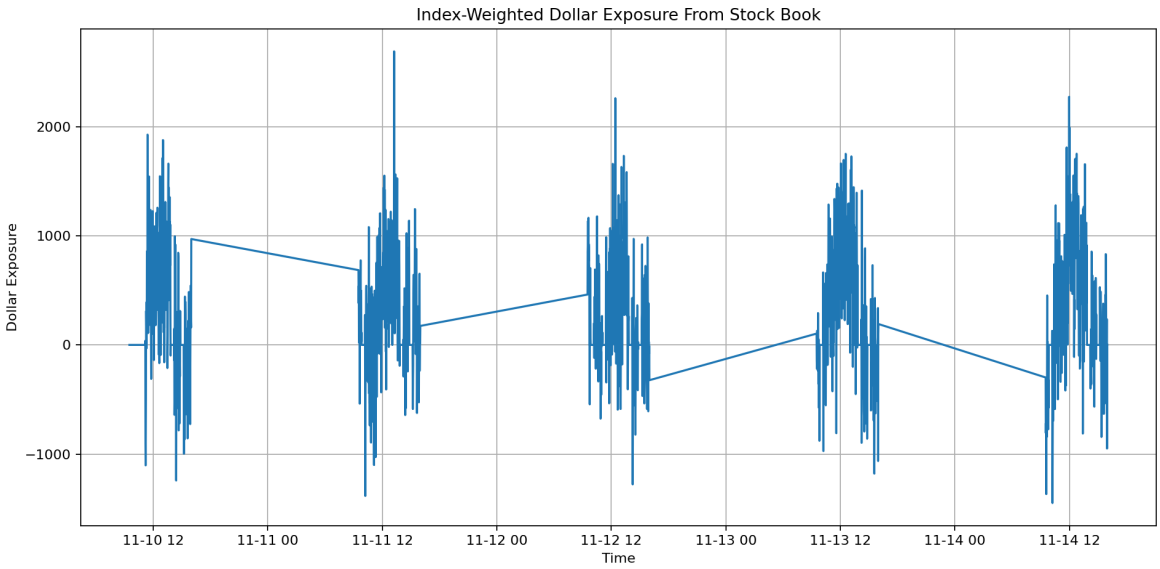

Random 16 tickers: Mid price (without last 4 minutes)
T0229
T0337
T0327
T0416
T0306
T0131
T0005
T0431
T0434
T0134
T0276
T0264
T0034
T0274
T0489
T0011

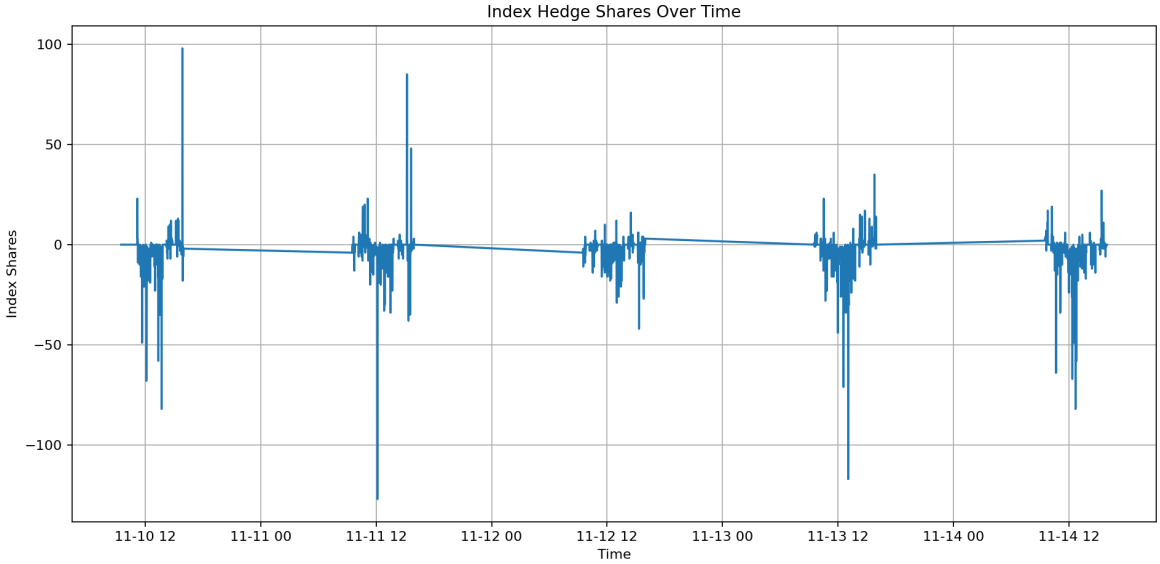

Random 16 tickers: Mid price during 2025-11-10
T0229
T0337
T0327
T0416
T0306
T0131
T0005
T0431
T0434
T0134
T0276
T0264
T0034
T0274
T0489
T0011

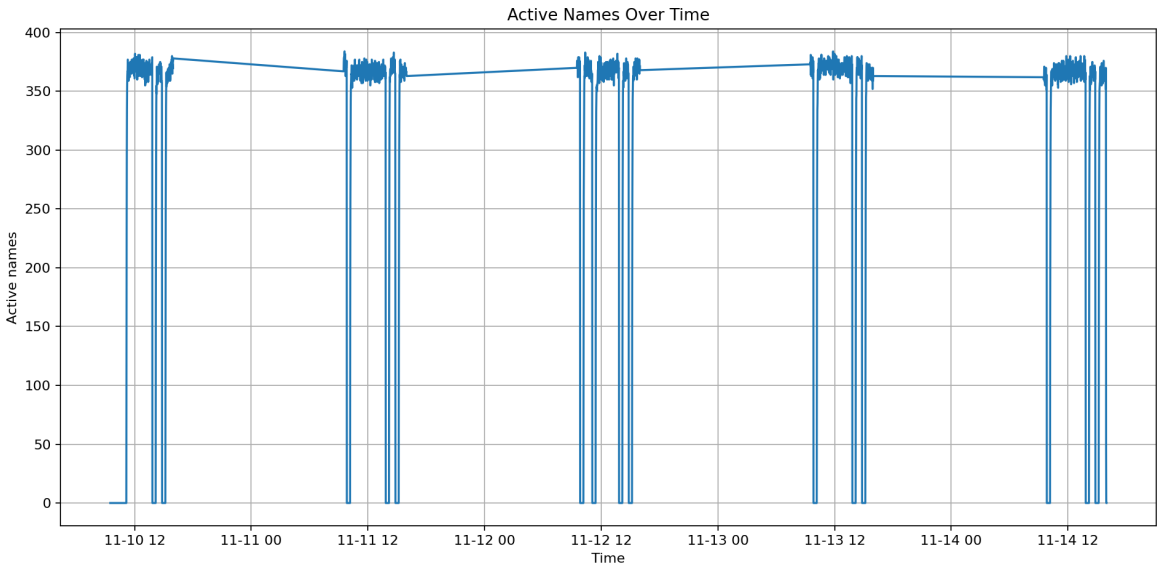

Distribution of means among tickers

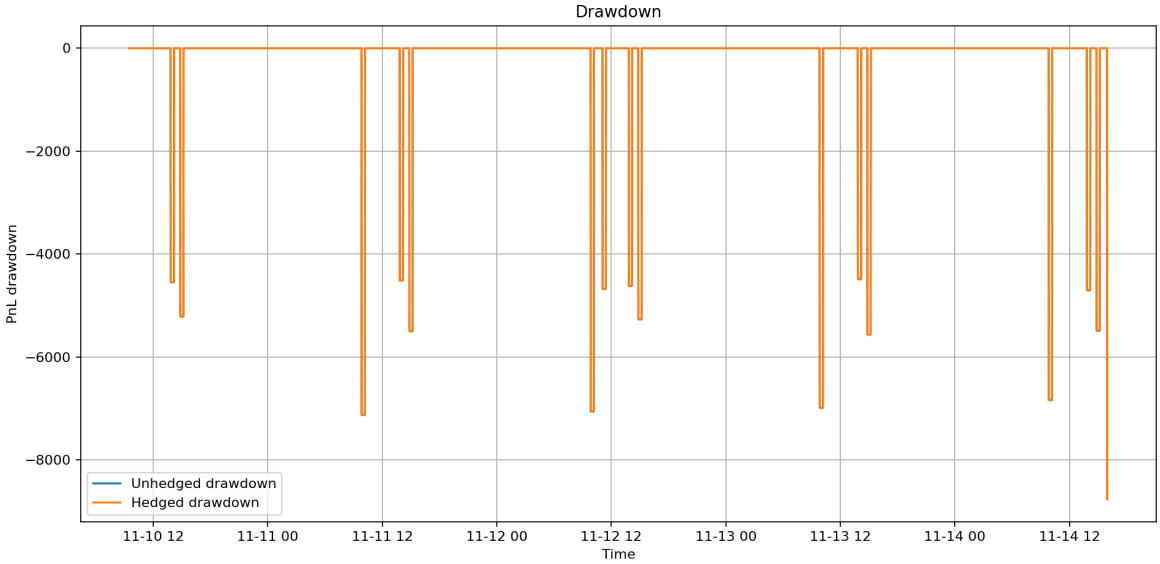

Distribution of standard deviations among tickers
140
120
100
80
60
40
20
0
112.0
112.2
112.4
112.6
112.8
113.0
113.2


Cumulative PnL
1e9
Gross Stock PnL
Net Stock PnL
Hedged PnL (Gross Hedge)
Hedged Net PnL
PnL
8
6
4
2
0
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12
Time


Index-Weighted Dollar Exposure From Stock Book
Dollar Exposure
2000
1000
0
−1000
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12
Time


Index Hedge Shares Over Time
Index Shares
100
50
0
−50
−100
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12
Time


Active Names Over Time
Active names
400
350
300
250
200
150
100
50
0
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12
Time


Drawdown
PnL drawdown
0
−2000
−4000
−6000
−8000
Unhedged drawdown
Hedged drawdown
11-10 12
11-11 00
11-11 12
11-12 00
11-12 12
11-13 00
11-13 12
11-14 00
11-14 12
Time